



*“Sablefish issues aren’t
just black and white”*



Sablefish Workshop Summary February 21-23, 2007 Seattle, Washington

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Introduction

In 1983 and 1993 sablefish (*Anoplopoma fimbria*) symposia were held that brought together scientists to share information and discuss future needs of sablefish research in the North Pacific. In 2004 a meeting was called at the Western Groundfish Conference to discuss the formation of a sablefish working group which would work to organize a third sablefish symposium. In the interim, an informal sablefish workshop was held February 21-23, 2007 at the Alaska Fisheries Science Center, Seattle. The purpose of the workshop was to bring together sablefish assessment scientists from the United States and Canada to exchange information, describe ongoing work, identify new avenues for research, and investigate cooperative research opportunities.

Twelve participants attended the meeting representing the Alaska Fisheries Science Center, Northwest Fisheries Science Center, Alaska Department of Fish and Game, Fisheries and Oceans Canada, Sigma Plus Consulting, and Simon Fraser University. Each agency provided general overviews of the sablefish stocks, fisheries, and management in

their respective jurisdictions. Roundtable discussions were held that focused on specific topics such as stock assessment, survey methodologies, life history studies, harvest strategy evaluations, and recruitment processes. What follows is a summary of the presentations and recommendations of the workshop participants for future research.

Fishery Management

Overview by Region

National Marine Fisheries Service - Alaska Federal (NMFS – Alaska)

The Alaska Fisheries Science Center (AFSC) presented an overview of sablefish fishery and management activities in the federally managed Alaskan exclusive economic zone (EEZ). Heavy fishing by foreign vessels during the 1970's led to a substantial decline in sablefish stocks in Alaska, which prompted fishery regulations and sharply reduced catches. The U.S. longline fishery began expanding in 1982 in the Gulf of Alaska; and in 1988 harvested all sablefish taken in Alaska, except minor joint venture catches.

The federally managed fishery in Alaska went to Individual Fishing Quota (IFQ) management in 1995. Quota shares were assigned initially to vessel owners or leaseholders who made at least one landing in the years 1988-1990. Each year, IFQs are assigned to individuals by multiplying the percentage of quota share they own by the annual harvest limit set for the sablefish fishery. Recent quotas have been near 20,000 mt. The majority of the sablefish catch comes from the eastern and central Gulf of Alaska, but the fishery also operates in the western Gulf of Alaska, Bering Sea, and Aleutian Islands. Fixed gear harvests approximately 85% of the sablefish quota and trawl gear approximately 15%. Pot fishing is banned in the GOA but is allowed in the Bering Sea and Aleutian Islands and accounts for nearly half of the IFQ catch in those areas. The sablefish fishery in Alaska is primarily a small boat fishery with nearly 400 vessels, and the season lasts from approximately March 1 – November 15. Fishery data is collected by fishery observers and through required and voluntary logbook programs. Harvests are strictly regulated and documented through IFQ landing reports and ADF&G fish tickets. Scientific data from the fishery is limited since fishery observers are not required on vessels under 60ft. Vessels over 60 ft are required to carry a fishery observer at least part of the time and are required to submit fishery logbooks. Data from vessels less than 60 ft come from landing tickets and a voluntary fishery logbook program.

The North Pacific Fishery Management Council (NPFMC) has established harvest rules that rely on biological reference points including fishing mortality and stock biomass levels to determine the status of the stock and subsequent harvest levels. Sablefish are managed under Tier 3 of NPFMC harvest rules, which assume reliable point estimates of fishing mortality and stock biomass levels are known. Each year a Stock Assessment and Fishery Evaluation report (SAFE) (<http://www.afsc.noaa.gov/refm/stocks/assessments.htm>) is prepared and reviewed for

sablefish under the Guidelines for Fishery Management Plans published by NMFS (<http://www.publicaffairs.noaa.gov/releases2005/jun05/noaa05-082.html>). This SAFE report summarizes biological and analytical results about sablefish stock status and includes acceptable biological catch (ABC) and overfishing level (OFL) recommendations for future years. The authors present their findings to the NPFMC Groundfish Plan Teams in September and November for initial review. Plan Team recommendations are then submitted to the NPFMC Scientific and Statistical Committee (SSC) for further review and final recommendations on ABC before final decisions on total allowable catches (TAC) and allocations are made by the NPFMC.

Many sablefish research activities are ongoing at the AFSC in addition to producing annual SAFE reports. A migration model is currently being updated to include recent tagging data and help address sablefish movement in Alaska and potentially help address quota apportionment by regions. Pot fishing has increased dramatically since 2000 in the Bering Sea and Aleutian Islands; and management and stock assessment concerns exist related to cannibalism in pots, escape ring sizes, soak times, and the effectiveness of biodegradable escape panels. Sablefish stomachs are currently being examined in the pot fishery for evidence of cannibalism. New research using satellite technology and physical oceanography are being investigated to better understand recruitment and help estimate the strength of year classes before they become available to the fishery. Parameters such as natural mortality, growth, and age at maturity are being investigated and values from older studies are being updated with new information. Finally, incorporating ecosystem considerations into stock assessments is continually evolving and improving our knowledge of sablefish and their interactions with other species and the environment.

Alaska Department of Fish and Game – State of Alaska, Southeast Alaska Sablefish (ADF&G)

The ADF&G presented an overview of the sablefish fishery and sablefish research activities for Southeast Alaska. There are two fisheries in the internal waters of southeast Alaska open to sablefish fishing, Chatham Strait and Clarence Strait. Both areas are managed as equal quota share fisheries, and both have a longline hook survey. The Chatham Strait area has the longer fishing history compared to Clarence Strait and receives far more research and management attention.

Commercial fishing has been documented in Chatham Strait since 1867. From 1985 to 1993 Chatham was managed under a limited entry program with quotas around 2.2 million round pounds; total catches ranged from 3 to 6 million round pounds. Since 1994 the Chatham fishery has been managed under a shared quota system. Adherence to quotas has improved; however quotas declined from about 4.1 million round pounds in 1994 to just over 2 million round pounds in 2006, in response to declining abundance.

Research and stock assessment activities began in 1980. The Department has conducted a longline survey for CPUE, a pot survey for releasing tagged fish, monitored CPUE and catch locations of the fishery, enumerated the abundance of marked and unmarked fish from fishery landings, and collected biological samples (length, weight, otoliths, sex, and

sexual maturity) from the fishery, longline survey, and pot survey. Since 2000, the gear configuration and soak time of the longline survey has been comparable to that of NMFS in the Gulf of Alaska. Since 1997, the Department has released fish with floy tags, PIT tags, sonic tags, and/or fin clips. Tagging was originally done on the longline survey, but due to concerns over hook-shyness, marked fish have been released from pot surveys since 2000.

The ADF&G manages all sablefish inside of the 3 mile state waters of Alaska as mandated by the Magnuson-Stevens Act. State sablefish fisheries occur in the Aleutian Islands, Cook Inlet, Prince William Sound, and Chatham Strait and Clarence Strait in Southeast Alaska. Ranges of catch in the different regions are between about 40,000 kg in Cook Inlet to one million kg in Chatham Strait. Longlining is the primary harvest method used in the state fishery, but there is some pot fishing in Clarence Strait and the Aleutian Islands. Fishery data comes from logbooks and landing reports.

Since statehood, the Chatham fishery has occurred in the late summer and fall. In 2004 and 2005 the Department allowed early season fishing in January, February and April. Information regarding length frequency, maturity and ages (not yet analyzed) was collected from commercial fishing vessels that fished in the off season and carried a Department observer aboard. The initial results of the sampling indicate that February is the peak of the spawning period, but that very few ripe females were captured in late January and early February. Additionally, a broader size distribution of fish was caught during these off season fishing events than were caught during the regular fishing season.

Currently ADF&G is collecting histological information for Chatham sablefish and will continue to collect samples during the winter spawning season. This will be used in part to attempt to validate macroscopic maturity assessment. Future work for the Department will also include trying to answer questions about spawning location, behavior, and timing. Detailed bathymetry and backscatter information collected by NOAA will be available to ADF&G for Chatham Strait at the end of 2007. With that information ADF&G will be able to better describe and quantify sablefish habitat in Chatham Strait.

Department of Fisheries and Oceans – British Columbia (DFO - BC)

Canadian sablefish are managed under the auspices of a collaborative agreement between Fisheries and Oceans Canada and the Canadian Sablefish Association. An Individual Vessel Quota (IVQ) management system was instituted in 1990 for the sablefish fishery where the directed catch is allocated in proportion to quota owned. There are forty-eight IVQ licenses in the directed fishery. An allocation of 8.75% of the total allowable catch accommodates sablefish bycatch in the multi-species trawl fishery. The fishery is managed as one geographic unit and occurs along the entire coastline, except that no directed fishery exists in coastal inlets. The directed sablefish fishery is primarily a longline trap (80%) and longline hook fishery. Average catch over the last 30 years is approximately 4,200 mt with a minimum size limit of 55 cm fork length for all gear sectors. Fishery data is collected through mandatory logbooks, observer coverage in the trawl fishery, and mandatory dockside validation for all sectors. Starting in 2006 there is

also mandatory at-sea camera coverage as a component of the Groundfish Integration Pilot Program. The Groundfish Integration Program is designed to provide comprehensive catch reporting by species and quota management through at-sea monitoring of catch, mandatory dockside validation, and the requirement to own or lease quota to cover the catch of all quota-managed groundfish species. Quota shares for species groups are attained by purchase or trading among license holders and can occur between gear sectors.

National Marine Fisheries Service – West Coast US (NMFS – West Coast)

The Northwest Fisheries Science Center assesses all sablefish in U.S. offshore waters from northern Washington to southern California. The sablefish stock size has decreased since the 1940's. Fishery landings in 1990 were about 10,000 mt but have decreased to about 5,500 mt. The three gear types participating in the sablefish fishery are trawl, hook-and-line, and pot gear. A coast-wide ABC exists, but there are tribal allocations as well as gear allocations (~45% trawl and ~45% fixed). The trawl fleet is managed primarily through bi-monthly trip limits whereas multiple regulations are used to manage the fixed gear fleet. Fishery data comes primarily from fishery observers and logbooks. Observer coverage for the trawl fishery is approximately 20-30% and for the hook-and-line fishery approximately 10%.

Overarching issues and recommendations

Whale depredation

Whale depredation is a major issue for the sablefish hook-and-line fishery in Alaska. Killer whales have depredated longline gear for many years in the Bering Sea, Aleutian Islands, western Gulf of Alaska, and Prince William Sound. Killer whale depredation is not prevalent in Chatham Strait or Frederick Sound. Sperm whale depredation has become very prevalent in the eastern and central Gulf of Alaska. Whale depredation in BC has not been reported as a major concern, possibly because the majority of fishing is done using trap gear and resident whales in B.C. may not have acquired the learned behavior. However, predation on newly released tagged fish by marine mammals has been observed. Current research regarding sperm whale depredation is being done by the AFSC and the SEASWAP project <http://www.seaswap.info/index.html> out of Sitka, Alaska.

Pot Regulations

Pot fishing in the Alaska federal IFQ fishery is banned in the Gulf of Alaska but has increased greatly since 2000 in the Bering Sea and Aleutian Islands. Recent attention has been given to pot fishing regulations in Alaska. Regulatory issues such as escape ring and bio-degradable escape panel requirements have been raised in addition to biological concerns such as cannibalism in pots. In 2006, the NPFMC requested that NMFS Alaska sablefish researchers conduct studies to address these issues. In British Columbia, pots

(more properly called traps) have been used for years in the sablefish fishery and are the primary gear type in the directed fishery. Pots have also been used for years in the U.S. West Coast fishery and are allowed in some ADF&G managed fisheries. The group discussed the pot fisheries in those areas and the range of regulations that have been applied.

DFO scientists noted that escape rings were introduced voluntarily by the industry in 1997 and by regulation in 1998 to increase the efficiency of their fishing operation, increase the size of retained fish, and promote stock health. The regulation specified that a 3-1/2 in diameter plug must be able to pass through the opening. Fishermen/DFO scientists noted fish of specific sizes being gilled in the escape rings. This phenomenon has initiated experimentation with escape ring shapes and construction on the part of the industry. Originally metal rings constructed from 3/16 to 1/4 inch diameter stainless steel stock were tied into the trap mesh to serve as escape rings. The twine resulted in a rough inside diameter and decreased the inside diameter of the ring; this may contribute to fish becoming gilled or reducing the size of fish that can escape via the rings. Alternative ring material such as a ring milled from 3/4 inch thick acrylic stock (like cutting board material) were tried, with a groove milled around the ring to enable attachment to the trap mesh. This provided a smooth inside surface and the increased width of material appears to reduce gilling. Openings made from polyethylene board with outer holes for rigging also provided a smooth opening and decreased gilling. Regardless of the design, escape rings with a minimum 3-1/2 inch diameter have been required by regulation since 1998, and their use was initiated, and strongly supported by the industry.

Biodegradable panels are also required by law in the Canadian sablefish fishery. Canadian researchers found that a rectangular panel with biodegradable twine on three sides was more effective for “rot” panels than a single slash sewn shut with biodegradable twine. Note that ADF&G also requires biodegradable panels to be used in all groundfish pots. ADF&G regulations require a single slash 18 inches in length in the pot sidewall that is laced with biodegradable cotton twine. The group agreed a biodegradable “rot” panel is likely an effective way to reduce the biological consequences of ghost fishing.

Workshop participants discussed previous experience with studies that examined the effectiveness of escape rings. Testing escape ring sizes, or effectiveness, is complicated because conducting experiments which enumerate the fish that escape from the traps is difficult. Canadian researchers have conducted escape ring studies but cautioned that objectives need to be clear and numerous replicates are necessary. To successfully measure the effectiveness of escape rings, methodologies must allow for all fish to be retained including those that exit the pot through the escape ring. The group also cautioned that possibly undesirable sex-specific selectivity may result from the use of escape rings because larger female sablefish are more likely to be retained.

The group agreed the potential for cannibalism of small fish in pots is low. Preliminary results from a study done in 2006 in the Bering Sea indicate that the minimum size of sablefish caught in pots was 40cm and that no cannibalism was recorded. Canadian

researchers concurred that despite not doing any directed cannibalism studies many stomach content analyses have been performed, and cannibalism has never been documented. Smaller sablefish (<40cm) are also rare in trap catches in British Columbia.

The working group lauded the initiative of the Canadian fishermen to take responsibility for such an aspect of their operation rather than require regulatory inducements to improve the benefits from the stock. The Canadian scientists agreed that the Canadian industry has shown pro-active behavior with respect to many aspects of the sablefish fishery.

Survey Descriptions/Issues

Overview by Region

NMFS - Alaska

Since 1978, the U. S. National Marine Fisheries Service (NMFS), Alaska Fisheries Science Center (AFSC) has conducted annual longline surveys with Japan (Japan-U.S. cooperative longline survey, 1978-94) and alone (1987-present, domestic longline survey). The survey has covered the upper continental slope (1978-present) and selected gullies (1987-present) of the Gulf of Alaska and the upper continental slope of the eastern Bering Sea (1982-94, biennially since 1997) and Aleutian Islands region (1980-94, biennially since 1996). A unique aspect of this survey is that the charter vessel retains most of the catch as compensation after the scientific data are recorded, which reduces government costs. The survey is conducted by catcher processors that retain the catch, while providing vessel, fuel, fishing crew, contract biologists, bait, and gear. NMFS provides the chief scientist and data processing equipment. It is a fixed station survey, covering the upper continental slope of the Gulf of Alaska annually and alternating each year between the Aleutian Islands and the Bering Sea. The survey is conducted each year by one vessel, occurs in the summer, and samples 87 stations taking approximately 3 months to complete. The survey objectives are to determine the relative abundance and size composition of commercially important species including sablefish, determine migration patterns of sablefish by tagging, and determine the age composition of sablefish through otoliths collections. The longline survey is the primary data input for the sablefish stock assessment used to manage the Alaskan fishery within the EEZ.

The survey subsamples for length, age, maturity, and sex composition and tags fish in each area surveyed. The current survey uses 2 m spaced circle hooks handbaited with squid. Catch rates are used to compute area weighted catch per unit effort values, which are treated as relative abundance indices in the stock assessment and area apportionments.

NMFS has also conducted a multispecies groundfish trawl survey every 2-3 years in the Aleutian Islands and the Gulf of Alaska. A trawl survey is conducted in the Bering Sea

shelf every year. This survey provides an absolute biomass index for each area and collects lengths and weights of sablefish. The stock assessment has recently begun incorporating shelf trawl survey information to supplement assessment of pre-recruit sablefish, which occur frequently in areas inshore of the longline survey area.

Survey concerns:

1. Killer and sperm whale depredation occurs on the survey gear. Killer whale depredation is easily detected and at current levels of depredation affected stations can be excluded from analyses. Sperm whale depredation is difficult to detect, but the effect on catch rates is estimated to be small.
2. Larger vessels, including those that conduct the survey, are changing to autobait systems. Our ability to maintain the same gear and handbaiting method in future surveys is a concern.
3. Hook competition is noted between sablefish and other species taken on the survey. Continued research is planned to evaluate the effect on the estimation of relative abundance from survey data.
4. Survey data collection improvements are being investigated, particularly progressing towards more electronic data collection.

ADF&G

A longline survey was started in the primary fishery area of Chatham Strait in 1988 and has continued to present. Survey stations were randomly placed and remain constant from year to year. The geographic coverage of the longline survey was increased in 1989 and 1997. Catch rates, age, size, sex, and maturity are obtained. Various changes to the survey have occurred over the years, such as soak time, vessel sources, longline type (snap vs. fixed), effort per station, and bait type (herring to squid). Longline survey CPUE is a relative index of abundance that has been used as an aid in quota setting and has also been an input of age structured models. The longline survey was also used to release marked fish from 1997-1999.

A pot survey was implemented in 2000 to release marked fish and has continued to present. Approximately 5000-7000 marked sablefish are released annually. Length measurements are collected on all fish. In addition, fish are subsampled for age, size, sex and maturity. Marked fish are recaptured during port sampling of commercial harvest. Mark-recapture or exploitation rate calculations based on tagged fish have provided the basis for the stock assessment since 2000.

Fisheries and Oceans Canada - BC

There are two surveys directed at sablefish that combine the functions of generating catch rate based abundance indices, release of tagged sablefish, and biological sampling. These surveys utilize sets of 25 traps with escape rings sewn shut. Outputs from these surveys are used for stock assessment.

1. Standardized survey with stock indexing, tagging and inlets components:

- Traditional standardized trap survey series from 1990 to 2006. The survey is conducted in 9 localities, with one 25 trap string per 100 fm depth interval (50 – 650 fm). Soak time is 24 hours using squid as bait.
- Traditional tagging survey re-initiated in 1991. This survey is conducted in October - November at 11 offshore localities with tagging goals of 300-1000 fish per area.
- Combined indexing and tagging sets in four mainland inlets. About half of the sablefish captured are tagged and released with the balance of the catch directed to biological samples.

2. Stratified random survey with stock indexing and tagging components:

- Stratified random trap survey to distribute tags randomly in 5 areas based on bathymetry and past fishing effort. Each area is stratified into 3 depth strata between 100–750 fms, with strata divided into 2 km by 2 km sampling units. In 2006, six replicates were achieved for each of the 15 combinations of area and depth strata. About 1/3 of the sablefish captures are tagged and released and 1/3 sacrificed for biological samples, with the balance surplus used for science needs.

Several multi-species groundfish trawl surveys have been initiated in the last three years. These surveys intercept sablefish but at this time it is not clear how useful they will be for stock assessment modeling. Divergence in catch rate time trends between the standardized survey and stratified random survey in the offshore zone suggest that the former may be sensitive to recruitment to the fishery.

NMFS–West Coast

The pot survey conducted since the early 1980's was discontinued in 1992. The current trawl survey began in the early 2000's to replace the previous triennial shelf survey and trawl slope survey conducted by the AFSC. The current survey is a stratified random block design with 15 minute tows. The survey was expanded to the Mexican border recently and is to be conducted annually depending on funding.

A recent pot study designed to estimate the proportion of population missed by restricting the trawl survey to 700 fathoms clearly indicated that the trawl survey is not reaching the full extent of sablefish population. Pot gear tended to catch more large sablefish than the trawl gear, due somewhat to availability differences as well as gear selectivity.

Survey concerns: Lack of a long term time series and limited coverage of sablefish population by current survey.

Stock assessment

Overview by region

NMFS – Alaska

The AFSC models the entire federally managed Alaska sablefish fishery as one population, integrating data from the Bering Sea, Aleutian Islands, and the Gulf of Alaska. This is done because analysis of tag recovery data has shown significant movement between areas implying that the existence of small, genetically unique stocks is unlikely. The model is sex-specific because males and females have different growth and maturity rates, and females attain higher maximum sizes. The model incorporates data from a variety of sources such as the historical Japanese longline survey and fisheries, the annual domestic NMFS longline survey, the biennial NMFS bottom trawl survey in the Gulf of Alaska, and the domestic fixed and trawl gear fisheries. The data provided by these sources include catch, relative abundance, age and length compositions, size-at-age, and maturity-at-age.

The model is a separable age-structured model coded in AD Model Builder, which uses a maximum likelihood and Bayesian approach to estimate parameters such as fishing mortality, absolute abundance, selectivity and catchability. The model is updated annually in conjunction with new longline survey and fishery data. The model is also used to project the population forward in time to calculate fishing mortality and quotas for subsequent years under various scenarios. These quotas are allocated to smaller management areas by computing a weighted 5-year average of survey and fishery catch rates by area. Bayesian MCMC methods are used to quantify the uncertainty of estimates. Current estimates show spawning biomass to be about 38% of unfished levels, which is close to the target of $B_{40\%}$.

ADF&G - Alaska

The ADF&G manages all sablefish fisheries inside of the 3-mile state waters boundary as mandated by the Magnuson-Stevens Act. Sablefish management is a mosaic of different levels of complexity and techniques in sub-regions in the State of Alaska. Regional catches range between about 40,000 kg in Cook Inlet to one million kg in Chatham Strait. Area quotas have been historically set using different methods. Several area quotas are set as a proportion of the overall NMFS quota or have no assessment. By far the largest level of harvest and biomass occurs in the Southeast Alaska Region. Therefore, this area is where the most state sablefish assessment research has been focused. The Department has explored a number of different methods for biomass estimation and forecasting. Methods have included the Petersen mark-recapture estimator, exploitation rate estimators, and age structured models. In the recent past, they have estimated current-year biomass with mark-recapture data and the Petersen estimator. To forecast, they divided the overall abundance into abundance-by-age-class using the age composition

from the longline survey, applied 10% mortality, assumed an equal number of age-4 recruits for the following year, and converted abundance to biomass using weight-at-age from the longline survey. To set annual quotas they applied an $F_{40\%}$ spawners-per-recruit harvest rate (same as AFSC) to the lower 90% confidence limit of the forecasted biomass. When these quotas are used for management, additional reductions are made to the recommended quota to account for bycatch from other fisheries and other estimated sources of mortality. The Department plans to continue mark-recapture studies. Despite difficulties with inconsistent age compositions over time and in accounting for fish movement in and out of Chatham Strait, the Department is revisiting the possibility of developing an age-structured assessment model for Chatham sablefish with the assistance of Dr. Franz Mueter.

DFO-BC

A wide variety of stock assessment models have been applied to BC sablefish data with 17 different models attempted over the last 20 years. In the late 1990s modeling complexity had progressed to an integrated mark-recapture age-structured model. Since that time the model was simplified to a monthly tagging model that incorporated several indices of stock abundance and was used to generate decision tables. The sablefish stock assessment team is now focused on evaluating management procedures and is conducting stakeholder consultation to refine fishery objectives. Unlike the situation in the U.S., there is no legislative enactment like the Magnuson-Stevens Act in Canada that requires fisheries be managed to maximum sustained yield.

NMFS - West Coast

The Northwest Fisheries Science Center assesses all sablefish in U.S. offshore waters from northern Washington to Southern California. The assessment is based on data from bottom trawl surveys, pot surveys, and fishery data. These data include catch and discard mortality, logbook catch-per-unit-effort, relative abundance, age-and-length compositions, size-at-age and maturity-at-age. The sablefish assessment for the West Coast is one of the few that incorporates an environmental index to aid in estimating recruitment. This index is based on Northern coastal sea level and Ekman transport. This index is incorporated in the stock-recruit relationship where high sea level predicts warmer water in-shore and lower recruitment.

The West Coast assessment is updated every three years, the last update was in 2005. The assessment currently uses Stock Synthesis II (SS2), an AD Model Builder based package produced by Rick Methot that allows the assessment scientist to choose from a wide array of data sources, functional relationships and graphical outputs. The outputs of the model indicate that stock abundance is relatively low. Parameters that are difficult to estimate in the model are catchability and steepness of the stock-recruitment relationship. A tendency for steepness to gravitate toward 0.2 is suspected to reflect a change in productivity that has resulted in its current depleted status. Movement of sablefish between areas is suspected, but only limited tagging data is available to evaluate this hypothesis. Therefore, a spatially explicit model is intractable at this time. Harvest rates

are computed at an F_{45} % SPR level, which is slightly more conservative than that used in Alaska. The SS2 package allows the use of prior distributions and Bayesian outputs such as the joint posterior distribution of natural mortality and catchability.

Overarching issues and recommendations

Model complexity and accessibility to stakeholders

The group recognized that stock assessments with more model complexity might work better under different management systems. In any system, the basic framework of the models needs to be understandable to all stakeholders involved. This does not preclude a complex age-structured assessment, but requires that the methods used be coherent, logical, and well documented so that stakeholders trust the results. In addition, different data sources are not always consistent. Complex models that integrate inconsistent data may be unstable due to a flat likelihood surface. The group noted that frequent changes to the structural components of the assessment model should be avoided to minimize confusion for fishery managers and stakeholders. While changing models is necessary when new concerns arise, in general a model should get to an acceptably reliable configuration and be applied over as long a period as possible. However, a proactive stock assessment scientist should make small, incremental improvements as new information comes to light.

Use of environmental indicators

The group acknowledged that using environmental data as a way to get upcoming recruitment information faster and more accurately is enticing, and the NMFS-West Coast efforts are laudable. However, the group recommended that any efforts to incorporate environmental indices into stock assessments should be preceded by a thorough understanding of biological mechanisms on recruitment, not strictly correlative success. In addition, it was proposed that any indices to be used in the stock assessment should be retrospectively tested to see if they would indeed result in more efficient management.

Communication and collaboration

The group indicated that much can be learned by meeting with other regions' assessment authors who often share similar or related issues. Some regions use quite different parameter estimates for life history quantities even though there may not be evidence to support that the fish are that different between areas. The group recommended that further collaboration concentrate on resolving or explaining these differences and the best ways to estimate certain parameters. The group also suggested that in the future, after data-sharing agreements have been put in place, a coast-wide assessment model might produce interesting and useful results.

Pot gear as an assessment concern

The group discussed the recent increase in pot gear in the Bering Sea and Aleutian Islands in Alaska. Assessment scientists agreed that the Alaska model is sufficiently complex, and that modeling pot gear and catch separately may lead to over-parameterization, given the marginal observer coverage on vessels with pot gear and that the model already estimates many selectivity curves.

Catch accounting and control rules

The group recognized that one of the most difficult issues for a stock assessment is the lack of good catch data. The group agreed that sablefish catch under IFQ systems are well estimated, but incidental catch and discards are less likely to be well measured. Once catch is well measured, then there must be a way to control catch to a level of biological sustainability. This should be done by having defined control rules to reduce fishing mortality when abundance appears to drop below some threshold. The group recommended that accounting for all fishing mortality, and having rules in place to control that mortality are the essential precursors to a successful stock assessment program.

Tagging, movement, and migration

Overview by region

NMFS – Alaska

Sablefish have been tagged by the Japanese and NMFS surveys in Alaska in large numbers since 1978, with the exception of 1994 and 1995 when no adult tags were released. A total of about 330,000 tags have been released, with recent releases averaging around 5,000 per year. Approximately 12,000 tags have been recovered as of 2005. Presently, about 600 tags per year are being recovered. These recoveries have been primarily in Alaskan waters, with a small percent recovered in Canada, and <1% recovered on the United States West Coast.

A paper by Heifetz and Fujioka (1991) used a Markovian movement model to estimate migration between large management areas and into Canada. Their general conclusion based on model estimates was to support previously reported information that small fish move west and then return as they get larger to spawning locations in the Eastern Gulf of Alaska.

Recently, this model was recoded into AD Model Builder, which will be more amenable to the geometric growth of data, be easier to test more complex stock hypotheses, and more accurately account for uncertainty. Preliminary results from this model were shown to the group and showed that the conclusions were similar, but lend less support for a strong westerly movement of small fish (e.g. they are just as likely to move east or west).

The model is capable of estimating movement rates to B.C., but does not have the required data to estimate rates from B.C. to Alaska.

Future plans are to attempt new movement hypotheses, with more dimensionality, and include smaller scale directional movement (e.g. inshore versus offshore or radial directionality). Eventually, this information may become important in Alaska for spatially-explicit stock assessment and apportionment of quotas.

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Future plans are to attempt new movement hypotheses, with more dimensionality, and include smaller scale directional movement (e.g., inshore versus offshore or radial directionality). Eventually, this information may become important in Alaska for spatially-explicit stock assessment and apportionment of quotas.

ADF&G –Alaska

Alaska Department of Fish and Game has been tagging sablefish in Chatham strait since the 1950's. ADF&G uses their tag-recovery data to both examine fish movement and to determine exploitation rates of the fishery. For a relatively small area, it was noted that there is a large amount of releases.

Recoveries from the 1970's and 1980's showed that roughly 12% of tags were found outside of Chatham Strait, while within the last five years only 1.5% have been found outside of Chatham Strait. NMFS data exist to show how many fish are moving into Chatham, but they have not been analyzed recently.

ADF&G presented posters that qualitatively indicated that many of the fish tagged in Chatham and Clarence Straits were recovered just on the BC side of the equidistant line between Canada and Alaska. In addition, these releases were recovered, albeit less frequently, further south along the B.C. continental slope.

DFO – BC

Department of Fisheries and Oceans in B.C. have done extensive tagging since 1977 with about 350,000 releases as of 2003. They are currently releasing about 15,000 tags per year, with high recovery rates of 2,000-5,000 per year. Reporting rates for tagged animals is presumed to be quite high.

Tagging data from fish tagged in inlets were presented, with a modeling procedure similar to Alaska that was used to estimate annual migration rates between areas. The preliminary results showed that inlet fish move extensively, with the majority going either offshore or to Alaskan waters. In addition, a few fish were recovered in the U.S. West Coast fisheries and Mexico, but most were intercepted in the southern British Columbia fishery. Data from offshore tagging were not presented, but it was estimated that about 80% of the offshore tags are recovered in the same general location. In general, inlet fish were smaller fish (<10 years old), while fish tagged offshore were larger fish (> 10 years old).

NMFS – West Coast

The West Coast has done a few isolated tagging experiments, but do not have an active tagging program. They do return recovered tags to NMFS – Alaska, ADF&G, and DFO – BC.

Overarching issues and recommendations

Data sharing

The group agreed that it would be worthwhile to pool resources when working on migration and tagging issues. Previous attempts at sharing tag data between NMFS and DFO have yielded limited benefits. However, renewed interest in tagging studies may provide an opportunity for mutually beneficial collaboration among agencies involved in sablefish. The group established some general guidelines that should be established prior to any new data sharing:

1. DFO, NMFS, and ADF&G all have rules governing data-sharing with other agencies, particularly for international cooperators. These protocols must be investigated and the limitations clearly outlined to collaborators.
2. All three agencies have rules governing the use of confidential fishery data, particularly when a location can be identified to one vessel, or one haul, which is the case for tag-recovery data. Adoption of an appropriate spatial scale for analyses could be established that will be useful for tagging models, but remain consistent with confidentiality guidelines.
3. There is a need to establish the objectives of tagging analyses prior to the exchange of tagging data. This step would identify the expected products from the arrangement and who (agencies and scientists) will participate in the collaboration;
4. Once data confidentiality and research objectives are established, some effort is required to explain to stakeholders how a data-exchange can be to their benefit. This may include identifying how the findings of research on tagging models can result in management benefits for a highly migratory species like sablefish.

The group identified several topics worthy of collaboration among agencies:

1. Exchanging physical tags among agencies and releasing them might be a way to see if reporting rates are different depending on the tagging method.
2. Since they are highly migratory, a coast-wide stock assessment model integrating data from the Bering Sea to the southern U.S. West Coast would be an interesting exercise.
3. Likewise, a coast-wide migration model would provide the clearest picture of the overall migration pattern of sablefish. NMFS – Alaska and DFO – BC have both made progress on migration models for their respective areas, but much more could be done with the combined data sets from the three agencies.

Life History Parameters

Age at 50% Maturity

Female age at 50% maturity estimates ranges from 3-4 years in BC to 7 years used by ADF&G. NMFS in Alaska currently uses an estimate of 6.5 years for females based on Sasaki's work in 1985. The group discussed several possible explanations for the different estimates and agreed differences may be attributable to artifacts of ageing techniques, which would make comparisons of maturity estimates difficult. Differences also exist in the staging criteria used in macroscopic evaluations of maturity, which may contribute to different age-at-maturity curves. Time of sampling was discussed. Past work has shown that summer samples are not ideal for determining maturity state. ADF&G and NMFS are currently doing histological work and collaboration was discussed. Ideally, fall and winter samples will be collected in 2007, which may provide updated maturity estimates based on histological methods rather than macroscopic evaluations.

Growth

Growth curves used by ADF&G and NMFS were shown and compared. Male growth curves are nearly identical but female curves differ between the agencies. Weight-at-length relationships between ADF&G-NMFS are very similar but weight-at-age relationships differ. Recommendations by the group were for ADF&G to look at the depth at which samples were collected because both NMFS and DFO provided data that indicate size can be a function of depth and that fish in deep water tend to be smaller at age than fish caught in shallower depths. Differences in growth relationships could also be attributed to differences in aging between agencies.

Natural Mortality

The following natural mortality (M) values are used:

NMFS – Alaska	0.10
ADF&G - Alaska	0.10
DFO – British Columbia	0.08

NMFS – West Coast 0.056

The different values used for natural mortality were discussed. The group noted many of the estimates were derived years ago and methodologies were not certain. One method relies on oldest aged fish, but estimates of age are dependent on the ageing techniques, which differ by region. Oldest ages range from 87 years in BC to 94 years in Alaska. It was agreed that the values used for natural mortality in complex models are not as crucial as some other parameters because natural mortality is often confounded with other parameters. This results in a trade-off against other parameters being estimated and therefore may be biased to achieve the best fit to the data. For simplistic forecasting methods, however, natural mortality values do make a considerable difference. No new studies are planned to address natural mortality but updating oldest age estimates for each agency may be worthwhile.

Life History

Juvenile sablefish (1-2 year olds) are commonly found in major inlets in BC and in small bays and inlets in southeast Alaska. Little is known about juveniles in western Alaska other than they are often reported near harbors and inshore areas when there are strong year classes.

Sablefish spawning habits and locations are uncertain. Older sablefish are usually considered to be most dense in Southeast Alaska, but are found throughout the Alaskan range. Recent data exploration has shown there may be a concentration of older sablefish in the western Aleutians. From analysis of data there does not appear to be any evidence of senescence in sablefish. Historical winter surveys found spawning fish in the mouth of Chatham Strait, an area where large older fish are commonly found. Recently, ADF&G conducted a winter fishery (January-April) to help define spawning areas, but ripe females were rarely caught which was interpreted as indicating either hook shyness or an absence of ripe females in those areas surveyed. In BC, peak catches occur in the winter and ripe females are commonly caught. Analysis of the winter BC fishery data may help define spawning habits of sablefish and possibly identify spawning aggregations.

Recruitment

Understanding recruitment processes for groundfish is considered to be a formidable task by most and an intractable task by others. The U.S. West Coast assessment has found some environmental proxies that are used in their stock assessment for helping predict recruitment (Schirripa and Colbert 2005). NMFS – Alaska presented some ideas on determining sablefish recruitment through environmental factors. The sources and spatial scales of data that are available for monitoring recruitment in Alaska were reviewed and include the following:

1. NMFS – ABL gillnet survey conducted on the annual longline survey from 1995-2004, which caught young of the year sablefish;

2. OCC/BASIS surface trawl survey from 2003-present catches juveniles (5-20cm) in the Bering Sea;
3. NMFS – biennial/triennial trawl surveys have caught larger (1+ yr olds) juveniles in different locations since 1977, and fish appear to move into the Bering Sea in large recruitment years;
4. Satellite data from NASA may provide reliable proxies of underlying processes to help reduce recruitment uncertainty;
5. Anecdotal reports of widespread abundance of age 1+ juveniles in nearshore waters should be documented as an indicator of exceptionally high egg/larval/YOY survival;
6. Recruitment is measured by the longline survey, but not until age 3-4 based on age compositions of the catch.

Evidence of young-of-the-year sablefish found on the BASIS survey in the Bering Sea suggests that pockets of spawning biomass in the Aleutians or Bering Sea Slope contribute more recruits than expected to the population. Jeff Fujioka suggested that the cumulative effects of trawling in the Eastern Bering Sea could have degraded the habitat for juvenile sablefish which used to be abundant there in the late 1970s according to trawl survey results.

NMFS – Alaska is exploring integrating some or all of the data sources listed above into a recruitment index to improve understanding of the pre-recruitment life history of sablefish. The group agreed that such knowledge would be useful and should be pursued. Some of the group suggested that if recruitment could be predicted with some accuracy several years earlier, this would promote more efficiency in setting quotas. Another question about recruitment to be answered is why stakeholders often see potentially large year classes disappear in the following several years of survey data. A mechanistic explanation of why some apparently strong year classes do not materialize in the fishery would be well received by stakeholders. The group concluded that detecting recruitments within the first few years is difficult, but remains as a research priority.

Ageing Session

Delsa Anderl, AFSC Ageing Lab, gave an excellent presentation on sablefish age determination and answered questions for workshop participants. Delsa provided a summary of the protocols in place for sablefish ageing at the AFSC, presented a summary and results of the inter-agency Committee of Age Reading Experts (CARE) sablefish otolith exchange, provided an overview of validation studies that have been done for sablefish, and provided examples illustrating the difficulties involved with sablefish ageing. AFSC protocol uses two experienced age readers who age approximately 2000 sablefish per year. One reader ages 100% of the samples and a second reader ages a 20% random sample of those. Since 2000, average agreement is 38.5% with a CV of 10.8%.

The CARE project has four participating agencies for sablefish and consistently exchanges sablefish otoliths. In 2007 each agency contributed 20-25 (N=89) non-

randomly selected otoliths, which are read by each group resulting in four sets of data. Percent agreement ranged from about 13% to 29% between agencies, and all groups were generally in agreement for fish under 15 years. Agencies are working together and are using CARE results to improve sablefish ageing techniques.

To date there have been six age validation studies for sablefish, and the AFSC is currently working on a study where otoliths are being read from known-age fish that were tagged beginning in 1985. This study is an extension of the work documented by Heifetz et al. (1999). Sablefish are a difficult species to age because they exhibit a broad “checky” pattern in early years, and the rules used for interpretation of patterns at the otolith edge differ from those applied to other species. These difficulties in pattern interpretation for sablefish lead to lumping or splitting issues in assigning fish age. Future studies for sablefish ageing include continuing the known age study and performing bomb C-14 validation of select otolith patterns to help validate the criteria used to interpret these patterns.

CARE information is available on the website: <http://www.psmfc.org/care/index.htm>

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